The Effect of Competition in Contests: A Unifying Approach*

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Abstract

We study all-pay contests under finite type-spaces, examining how the competitiveness of the prize structure affects effort. We characterize the unique symmetric equilibrium and analyze it through a novel quantile representation. Our results establish the most competitive winner-takes-all contest as robustly optimal—it maximizes the total effort across the top q agents, for any q, under linear, concave, and even moderately convex costs. We conjecture that the degree of convexity required to overturn the optimality of winner-takes-all shrinks as the environment approaches complete information, suggesting a reconciliation of the contrasting results in the complete-information (Fang, Noe, and Strack, 2020) and continuum type-space (Moldovanu and Sela, 2001) environments. While winner-takes-all is optimal, the effect of competition is not monotone, as we uncover an *interior discouragement effect*: shifting value toward better-ranked intermediate prizes may reduce effort when inefficient types are relatively likely. An experiment provides qualitative support for these findings.

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1 Introduction

A central question in contest theory is how prize structures shape agents' incentives to exert effort, and in particular, which structures maximize effort. The answer turns out to depend critically on the information environment. Under complete information, the prize structure is irrelevant with linear costs (Barut and Kovenock, 1998), and minimally competitive structures with multiple prizes are optimal when costs are convex (Fang, Noe, and Strack, 2020). By contrast, with a continuum of types, the most competitive winner-takes-all structure is optimal under linear costs (Moldovanu and Sela, 2001), and can remain so even under convex costs (Zhang, 2024). Yet, what drives this divergence remains unclear, and it is an open question which (if any) of these findings extend to the intermediate and fundamental case of a finite type-space. This gap was also noted in a survey article by Sisak (2009), who conjectured that multiple prizes might be optimal:

> *"The case of asymmetric individuals, where types are private information but drawn from discrete, identical or maybe even different distributions, has not been addressed so far. From the results ..., one could conjecture that multiple prizes might be optimal even with linear costs."*

In this paper, we address this question by studying all-pay contests where ex-ante symmetric agents have private abilities drawn from a finite type-space. Given a contest—an assignment of prize values to ranks—and their private types, agents simultaneously choose effort levels, incur the associated type-dependent costs, and receive prizes based on the rank order of their efforts. The finite type-space framework embeds complete information as a special case and can approximate any continuum type-space. Thus, our analysis not only bridges a gap in the literature, but provides a unifying approach offering insights into the contrasting results in these extreme environments. Beyond its theoretical appeal, the framework enables theoretically founded experimental investigation.

We begin by characterizing the unique symmetric equilibrium of the Bayesian game. We show that equilibrium is in mixed strategies and exhibits a monotonic structure: different types randomize over disjoint but contiguous intervals, with more efficient types always outperforming less efficient ones. To overcome the analytical complexity of this mixed equilibrium, we introduce a quantile representation: the quantile function of the ex-ante equilibrium effort distribution, wherein the quantile level corresponds to the probability of outperforming an arbitrary opponent. The representation recasts the mixed equilibrium as a single deterministic effort schedule over quantile levels, and—since the quantile level is ex-ante uniformly distributed—delivers a tractable expression for expected effort. In a sense, it extends the assortative-allocation ideas used in continuum type-space environments to the mixed equilibria under finite types, in a way that remains useful even under complete information, where assortative allocations do not apply.

We use this representation to study how increasing competition—shifting value from lower-ranked prizes to better-ranked prizes—affects equilibrium effort. We first characterize the effects on expected effort under linear costs. By a relabeling argument, equilibrium effort under linear costs coincides with equilibrium effort *costs* under an arbitrary cost function, and these effects carry over directly to expected effort costs. We then identify conditions under which they extend qualitatively to expected effort itself, and to related objectives.

These results establish the winner-takes-all prize structure as robustly optimal: it maximizes the total effort of the top q agents, for any q, under linear, concave, and even moderately convex costs. Intuitively, given an arbitrary prize structure, shifting value to the best prize strengthens incentives for the most efficient types, who are cheapest to incentivize. Under linear costs, this encouragement effect more than compensates for the discouragement induced among less efficient types. Under concave costs, the encouragement is further amplified, as marginal costs are lower at higher effort levels. Under convex costs, by contrast, inducing additional effort at high levels becomes progressively more expensive; nevertheless, for a parametric family of cost functions, we show that as long as costs are not too convex, the encouragement effect continues to outweigh the rising marginal costs. These forces are only strengthened when the designer cares about the effort of the top few agents, since such objectives place greater weight on precisely the types that winner-takes-all incentivizes most. Our results thus extend the optimality of winner-takes-all, established in continuum type-space models (Moldovanu and Sela, 2001), to finite type-spaces and to more general objectives, and resolve Sisak (2009)'s conjecture in the negative for the case of identical distributions.

Our analysis further suggests a reconciliation of the contrasting results in the complete-information and continuum type-space environments. Recall that for complete information, the prize structure is irrelevant under linear costs and winner-takes-all actually minimizes effort under convex costs. But as soon as we move away from complete information, winner-takes-all is optimal with linear and even moderately convex costs. These findings and our analysis lead us to conjecture that the degree of convexity required to overturn the optimality of winner-takes-all shrinks as the environment approaches complete information. While we do not offer formal results along these lines, we illustrate in a binary types environment that the encouraging effect of shifting value to the best prize under linear costs is single-peaked in the probability of the inefficient type, vanishing at both extremes as the environment approaches complete information. In this sense, the contrasting results in the two extreme environments emerge as limiting cases of a single comparative static in the amount of type uncertainty.

Despite the robust optimality of the most competitive winner-takes-all contest, the effect of competition is not monotonic. Specifically, it is not generally the case that shifting value toward better-ranked prizes increases effort. We identify an *interior discouragement effect*: shifting value from lower-ranked prizes to better-ranked intermediate (but not top) prizes may reduce aggregate effort when inefficient types are sufficiently likely. Intuitively, such transformations discourage inefficient types—much like shifting value to the top prize—but the encouraging effect on efficient types is muted, relative to the top prize. This is because the transformation also dilutes the incentive to compete for the best prize. Consequently, the encouragement effect on the most efficient types is dampened, and their effort may even decline. When inefficient types are sufficiently prevalent, the discouragement effect dominates and total effort falls. This non-monotonicity has implications for the design problem in cases where the top prize may be capped due to institutional or fairness reasons.

Our finite type-space framework enables direct experimental testing of the model's predictions. Since implementing a continuum of types is infeasible, prior experiments have relied on large but finite type-spaces, assuming equilibrium properties extend from the continuum setting (Müller and Schotter, 2010).[1] While our equilibrium convergence result formally justifies this assumption, our framework requires no such approximation: equilibrium predictions are exact for any finite type-space. Our experiment employs a simple binary type-space with linear costs in which the inefficient type is relatively likely—the environment where the interior discouragement effect is predicted to arise. We vary the prize structure

[1] Rasooly and Gavidia-Calderon (2020); Swarthout and Walker (2009) illustrate how the set of equilibria can differ qualitatively between finite and continuous environments in the all-pay auction and the Groves–Ledyard mechanism, respectively.

by gradually increasing competitiveness across four contests, including winner-takes-all. The results reveal an over-provision of effort, particularly by the inefficient type, but aggregate patterns broadly align with our comparative statics: winner-takes-all remains optimal, and the interior discouragement effect receives partial support. Specifically, although efforts do not decline as theory predicts, we observe no significant increase in effort when competition intensifies in the interior.

Related literature

The existing game-theoretic literature in contests has predominantly focused on the design problem in environments where the type-space is either a continuum or a singleton (the complete information case). The results highlight how the structure of the optimal contest can vary significantly depending on the environment. For the continuum type-space, the most competitive winner-takes-all contest has been shown to be optimal under linear or concave costs (Moldovanu and Sela, 2001), in some cases under convex costs (Zhang, 2024), with negative prizes (Liu, Lu, Wang, and Zhang, 2018), and under general architectures (Moldovanu and Sela, 2006; Liu and Lu, 2014). In comparison, in complete information environments, the minimally competitive budget allocation, wherein all agents but one receive an equal positive prize, has been shown to be a feature of the optimal contest quite generally (Barut and Kovenock, 1998; Letina, Liu, and Netzer, 2020, 2023; Xiao, 2018). In a general large-contest framework, Olszewski and Siegel (2016, 2020) show that awarding multiple prizes of descending sizes is optimal under convex costs. Other closely related work has examined how competition affects effort in the complete information setting (Fang, Noe, and Strack, 2020), and in the continuum type-space setting (Goel, 2025; Krishna, Lychagin, Olszewski, Siegel, and Tergiman, 2025).[2]

There is a related literature on contests with a finite type-space, much of which assumes binary type-spaces or a small number of agents and focuses on characterizing equilibrium properties under correlated or asymmetric types. Siegel (2014) establishes the existence of

[2] In early work, Glazer and Hassin (1988) highlight the distinction between the two environments by solving the problem in some special cases. Other related studies include Schweinzer and Segev (2012); Drugov and Ryvkin (2020), who examine the budget allocation problem under different contest success functions. For general surveys of the literature in contest theory, see Corchón (2007); Sisak (2009); Konrad (2009); Vojnović (2015); Fu and Wu (2019); Chowdhury, Esteve-González, and Mukherjee (2023); Beviá and Corchón (2024).

a unique equilibrium under general distributional assumptions. With correlated types, Liu and Chen (2016) show that the symmetric equilibrium may be non-monotonic when the degree of absolute correlation is high, Rentschler and Turocy (2016) highlight the possibility of allocative inefficiency in equilibrium, while Tang, Fu, and Wu (2023) and Kuang, Zhao, and Zheng (2024) explore the impact of reservation prices and information disclosure policies, respectively. With asymmetric type distributions, Szech (2011) shows that agents may benefit from revealing partial information about their private types, while Chen (2021) characterizes equilibrium outcomes for varying levels of signal informativeness.[3]

There is also a significant literature studying contests through incentivized laboratory experiments. Perhaps the most closely related work is Müller and Schotter (2010), who provide evidence broadly consistent with the predictions of Moldovanu and Sela (2001): winner-takes-all is optimal under linear costs, whereas splitting prizes is favored under convex costs. Two other studies, Barut, Kovenock, and Noussair (2002) and Noussair and Silver (2006), examine all-pay auctions with private valuations, a setting strategically equivalent to all-pay contests with private costs. All three experiments employ large finite type-spaces to approximate continuum-type equilibria, an assumption that our convergence result formally justifies. Like us, these studies observe substantial overbidding relative to theoretical benchmarks. A key difference, however, is that the overbidding there comes largely from the more efficient types, whereas in our experiment it is the inefficient types who significantly overbid, accounting for much of the aggregate over-provision of effort.[4]

The paper proceeds as follows. Section 2 presents the model. Section 3 characterizes the symmetric equilibrium and develops the quantile representation. Section 4 studies the effect of competition under linear costs, and Section 5 extends the analysis to general cost functions. Section 6 presents the experiment, and Section 7 concludes. The equilibrium convergence result is developed in Appendix D.

[3]Other related work in contests considers imperfectly discriminating contests (Ewerhart and Quartieri, 2020), altruistic or envious types (Konrad, 2004), and asymmetric information (Einy, Goswami, Haimanko, Orzach, and Sela, 2017). There is also a related literature on mechanism and auction design with finite type-spaces (Maskin and Riley, 1985; Jeong and Pycia, 2023; Vohra, 2012; Lovejoy, 2006; Doni and Menicucci, 2013; Elkind, 2007).

[4]In other related work, Brookins and Ryvkin (2014) compares behavior under complete and incomplete information in Tullock contests. See Dechenaux, Kovenock, and Sheremeta (2015) for a survey of the experimental literature on contests.

2 Model

Game

A *contest environment* is a tuple $(N+1, \Theta, p)$, where

- $N+1$ is the number of agents,
- $\Theta = \{\theta_1, \ldots, \theta_K\}$ is a finite set of types, with $\theta_1 > \cdots > \theta_K > 0$, and
- $p = (p_1, \ldots, p_K)$ is a probability distribution over Θ, with $p_k > 0$ for all k.

Each of the $N+1$ agents has a private type $\theta \in \Theta$, which represents the agent's marginal cost of exerting effort. Thus, lower values of θ correspond to more efficient types, with θ_1 being the least efficient type and θ_K the most efficient. Types are drawn independently according to p. We let $P_k = p_1 + \cdots + p_k$.

A *contest* $v = (v_0, \ldots, v_N)$ assigns a prize value to each rank, with $v_0 \leq \cdots \leq v_N$ and $v_0 < v_N$.

Given a contest environment $(N+1, \Theta, p)$, contest v, and their private types, agents simultaneously choose effort. They are ranked according to their efforts, with ties broken uniformly at random, and awarded the corresponding prizes. An agent who outperforms exactly $m \in \{0, \ldots, N\}$ of the N other agents is awarded prize v_m. If an agent of type $\theta \in \Theta$ wins prize v_m after exerting effort $x \geq 0$, their vNM utility is

$$v_m - \theta x.$$

We focus on linear effort costs for now. In Section 5, we extend the model to general cost functions $c(\cdot)$, where we show that the equilibrium under linear costs more generally characterizes equilibrium *effort costs* under arbitrary c, and identify conditions under which the effect on effort costs is informative about the effect on effort itself.

The Bayesian game induced by v is strategically equivalent to the game induced by the contest w, where $w_m = v_m - v_0$ for all $m \in \{0, \ldots, N\}$. We therefore normalize $v_0 = 0$ and focus on contests in the set

$$\mathcal{V} = \{v \in \mathbb{R}^{N+1} : 0 = v_0 \leq v_1 \leq \cdots \leq v_N,\ v_N > 0\}.$$

Equilibrium

We focus on symmetric Bayes-Nash equilibria, as is standard in this literature.[5] A symmetric Bayes-Nash equilibrium is a strategy profile in which all agents use the same strategy—a mapping from types to (possibly random) effort—such that, for every agent and every type $\theta \in \Theta$, the prescribed effort maximizes expected utility given that all other agents follow the same strategy. We denote a symmetric Bayes-Nash equilibrium by $(X_1, X_2, \ldots, X_K)$, where $X_k \sim F_k$ is the effort exerted by an agent of type θ_k. We further denote by $X \sim F$ the ex-ante equilibrium effort of an arbitrary agent, so that for any $x \in \mathbb{R}$, $F(x) = \sum_{k=1}^{K} p_k F_k(x)$. Accordingly, the expected equilibrium effort of an arbitrary agent is

$$\mathbb{E}[X] = \sum_{k=1}^{K} p_k \mathbb{E}[X_k].$$

Competition

We are interested in examining how increasing competitiveness of a contest influences equilibrium effort. Following the literature (Fang, Noe, and Strack, 2020; Goel, 2025), a contest $v \in \mathcal{V}$ is *more competitive* than $w \in \mathcal{V}$ if the prizes are more unequal, measured using the Lorenz order, i.e.,

$$\sum_{i=0}^{m} v_i \leq \sum_{i=0}^{m} w_i \text{ for all } m \in \{0, 1, \ldots, N\},$$

with equality for $m = N$.

Importantly, if v is more competitive than w, v can be obtained from w through a sequence of transfers from lower-ranked prizes to higher-ranked prizes. The marginal effect of such a transfer—from prize m' to prize m, with $m > m'$—on expected equilibrium effort is captured by

$$\frac{\partial \mathbb{E}[X]}{\partial v_m} - \frac{\partial \mathbb{E}[X]}{\partial v_{m'}}.$$

Our objective is to understand how this effect may depend on the underlying contest environment, the structure of the contest, as well as the specific pair of prizes.

[5] Asymmetric equilibria may exist in certain instances of the model. For example, Barut and Kovenock (1998) characterize equilibria in the complete-information case ($K = 1$), which includes asymmetric equilibria when the contest $v \in \mathcal{V}$ is such that the bottom two prizes are identical, i.e., $v_1 = v_0 = 0$.

We further discuss implications for the classical design problem of allocating a fixed budget $V \in \mathbb{R}_+$ across prizes to maximize effort. Notice that among all contests in $\mathcal{V}$ that distribute the entire budget, the winner-takes-all contest $v = (0, 0, \ldots, 0, V)$ is the most competitive, while the contest that allocates the budget equally among all but the worst-performing agent $v = (0, \frac{V}{N}, \ldots, \frac{V}{N})$ is the least competitive.

Notation

Suppose an agent outperforms each of the other N agents independently with probability $t \in [0, 1]$. Let

$$H_m^N(t) = \binom{N}{m} t^m (1-t)^{N-m}$$

denote the probability that the agent outperforms exactly m out of N agents. Define

$$H_{\leq m}^N(t) = \sum_{i=0}^{m} H_i^N(t) \text{ and } H_{\geq m}^N(t) = \sum_{i=m}^{N} H_i^N(t),$$

as the probabilities that the agent outperforms at most m and at least m agents, respectively. Finally, given a contest $v \in \mathcal{V}$, define

$$\pi_v(t) = \sum_{m=0}^{N} v_m H_m^N(t),$$

to be the expected prize received by the agent under v.

3 Equilibrium

In this section, we characterize the symmetric Bayes–Nash equilibrium of the Bayesian game. We derive necessary conditions that uniquely pin it down, verify sufficiency, and develop a useful representation.

To begin, we establish a robust structural property: for any contest environment $(N + 1, \Theta, p)$ and contest $v \in \mathcal{V}$, in equilibrium, different agent types mix over contiguous intervals, with more efficient types choosing higher effort than less efficient ones.[6]

[6] Wang (1991) studies common-value auctions with finite signals and finds a similar equilibrium structure. Related interval structures of mixed-strategy equilibria appear in two-player all-pay auctions with finite correlated types (Siegel, 2014; Rentschler and Turocy, 2016).

Lemma 1 (Equilibrium structure)**.** *For any contest environment* $(N+1, \Theta, p)$ *and contest* $v \in \mathcal{V}$*, if* $(X_1, \ldots, X_K)$ *is an equilibrium, then there exist boundary points*

$$0 = b_0 < b_1 < \cdots < b_K$$

such that, for each k*,* X_k *is continuously distributed on* $[b_{k-1}, b_k]$*.*

In words, agents of the least efficient type θ_1 mix over the interval $[0, b_1]$, agents of type θ_2 mix over $[b_1, b_2]$, and so on, up to agents of the most efficient type θ_K, who mix over $[b_{K-1}, b_K]$. The supports are disjoint across types because moving from one effort level to another generates the same change in expected prize for all types, while the change in cost is type-dependent. Consequently, two distinct types cannot both be indifferent between the same pair of effort levels, implying that their supports can intersect only at boundary points. In particular, more efficient agents always outperform less efficient agents, and mixing arises solely because agents may compete against others of the same type.

The mixing distributions, $F_1, \ldots, F_K$, must satisfy the indifference conditions. From Lemma 1, if a type-θ_k agent chooses effort $x_k \in [b_{k-1}, b_k]$, it outperforms an arbitrary agent with probability

$$P_{k-1} + p_k F_k(x_k).$$

Since the agent must be indifferent across all such effort levels, the equilibrium distribution F_k must satisfy, for all $x_k \in [b_{k-1}, b_k]$,

$$\pi_v\big(P_{k-1} + p_k F_k(x_k)\big) - \theta_k x_k = u_k, \tag{1}$$

where u_k denotes the equilibrium utility of type θ_k. Given the equilibrium utilities (and the boundary points), since π_v is strictly increasing on $(0, 1)$, Equation (1) uniquely pins down F_k.

It remains to determine the equilibrium utilities. Note that u_k equals the payoff of type-θ_k agent at $x_k = b_{k-1}$, and evaluating Equation (1) at $x_k = b_k$ (where $F_k(b_k) = 1$) yields $b_k = \frac{\pi_v(P_k) - u_k}{\theta_k}$. The following iterative argument, initiated by $b_0 = 0$, then gives the equilibrium utilities:

$$
\begin{aligned}
b_0 = 0 \Rightarrow u_1 &= 0 && \text{(type-}\theta_1\text{ agent's utility at } b_0\text{)} \\
\Rightarrow\ b_1 &= \frac{\pi_v(P_1) - u_1}{\theta_1} && \text{(from Equation (1) using } u_1\text{)} \\
\Rightarrow\ u_2 &= \pi_v(P_1) - \theta_2 \cdot b_1 && \text{(type-}\theta_2\text{ agent's utility at } b_1\text{)} \\
\Rightarrow\ b_2 &= \frac{\pi_v(P_2) - u_2}{\theta_2} && \text{(from Equation (1) using } u_2\text{)} \\
\Rightarrow\ u_3 &= \dots
\end{aligned}
$$

Proceeding in this way, we can solve explicitly for the equilibrium utilities, and Equation (1) then uniquely determines the candidate equilibrium; we verify in the proof of Theorem 1 that it is indeed an equilibrium. However, the implicit form of Equation (1) makes the equilibrium difficult to work with directly.

We develop an alternative representation—using the *quantile function*—that renders the equilibrium analytically tractable. Given a symmetric equilibrium $(X_1, X_2, \dots, X_K)$ with ex-ante effort distribution F, let

$$X_v : [0,1] \to \mathbb{R}_+$$

denote the quantile function of F, assigning to each quantile level $t \in [0,1]$ the corresponding equilibrium effort. The representation carries a natural interpretation in our context: since an agent exerting effort $X_v(t)$ outperforms an arbitrary opponent with probability exactly t, the quantile level t is the probability of outperforming an arbitrary agent. In effect, the representation identifies each agent with a quantile level $t \in [0,1]$: rather than tracking K types, each mixing over an interval of efforts, the equilibrium is summarized by a single deterministic effort schedule over a continuum of quantile levels.

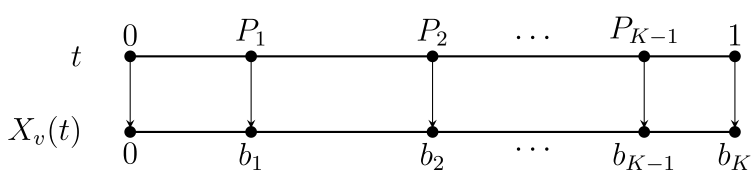


Figure 1: The quantile representation of the symmetric equilibrium.

From Lemma 1, quantile level $t = 0$ corresponds to zero effort, $t = 1$ corresponds to maximal effort b_K, and $t = P_k$ maps to the boundary effort b_k. This is illustrated in Figure 1.

In general, the indifference condition in Equation (1) can be restated directly in terms of this representation: the equilibrium effort schedule must satisfy, for all $t \in [P_{k-1}, P_k]$,

$$\pi_v(t) - \theta_k X_v(t) = u_k. \tag{2}$$

This leads to the following characterization of symmetric equilibria.

Theorem 1 (Equilibrium characterization)**.** *For any contest environment $(N+1, \Theta, p)$ and contest $v \in \mathcal{V}$, there exists a unique symmetric equilibrium $(X_1, \ldots, X_K)$. It is characterized by the quantile function $X_v : [0,1] \to \mathbb{R}_+$ of the ex-ante effort distribution F: for each $t \in [0,1]$,*

$$X_v(t) = \frac{\pi_v(t) - u_{k(t)}}{\theta_{k(t)}}, \tag{3}$$

where $k(t) = \max\{k : P_{k-1} \leq t\}$, and the equilibrium utilities are given by

$$u_k = \theta_k \left[\sum_{j=1}^{k-1} \pi_v(P_j) \left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j} \right) \right]. \tag{4}$$

This quantile representation of symmetric equilibrium—formulated in terms of the probability of outperforming an arbitrary agent—is central to our subsequent analysis. In a sense, it extends the assortative allocation idea typically used to analyze pure-strategy equilibrium in continuum-type environments to the mixed-strategy equilibrium that arises in our setting with finite types, in a way that remains useful even in the complete-information case, where assortative allocations do not apply. We conclude this section with two remarks, describing how our equilibrium characterization relates to those in the complete-information and continuum type-space environments, and how it extends to richer non-parametric type-spaces consisting of arbitrary but ordered cost functions.

Remark 1 (Equilibrium convergence)**.** *The equilibrium under the finite-type space framework exhibits both the mixed structure characteristic of complete information environments (Barut and Kovenock, 1998) and the monotonic structure observed in environments with a continuum of types (Moldovanu and Sela, 2001). The complete information environment is clearly a special case of our model. We also establish an equilibrium convergence result for the continuum type-space environment (Theorem 5 in Appendix D), which implies that the (pure-strategy) equilibrium in any continuum type-space can be well-approximated by the equilibrium of a sufficiently large and appropriately chosen finite type-space. Intuitively, as the finite type-space becomes large, the interval over which an agent of a certain type mixes* *shrinks, and essentially converges to the effort level prescribed by the pure-strategy equilibrium under the continuum type-space. The representation thus provides a single formulation that spans complete information, finite types, and the continuum limit.*

Remark 2 (Non-parametric type-spaces)**.** *The equilibrium characterization in Theorem 1 extends to environments in which the type space is not parametric, but consists of cost functions that are ordered. Specifically, let $\mathcal{C} = \{c_1, \ldots, c_K\}$ be the type-space containing cost functions such that for each $k \in [K]$, $c_k(0) = 0$, c_k is strictly increasing and differentiable on $(0, \infty)$, and $\lim_{x\to\infty} c_k(x) = \infty$. Suppose further that types are ordered by marginal costs: for every $x > 0$,*

$$c'_1(x) > c'_2(x) > \cdots > c'_K(x).$$

In this more general framework, there is a unique symmetric equilibrium in which different agent types mix over contiguous intervals (ordered by type). It is characterized by the quantile function of the ex-ante effort distribution: for each $t \in [0, 1]$,

$$X_v(t) = c^{-1}_{k(t)}\big(\pi_v(t) - u_{k(t)}\big),$$

where the equilibrium utilities are determined by the same iterative argument as above.

4 Competition

In this section, we examine how increasing competitiveness of a contest, by shifting value to better prizes, influences expected equilibrium effort.

To begin, the equilibrium characterization in Theorem 1 yields a tractable expression for expected effort. Since X_v is the quantile function of the ex-ante effort distribution F, and the quantile level $F(X)$ of a draw $X \sim F$ is uniformly distributed on $[0, 1]$ (the probability integral transform), the expected equilibrium effort is

$$\mathbb{E}[X] = \int_0^1 X_v(t)dt.$$

From Equation (3), $X_v(t)$ is linear in prize values, which implies that expected effort is also linear in prize values. Solving the integral yields the required coefficients.

Lemma 2 (Expected effort)**.** *For any contest environment* $(N+1, \Theta, p)$ *and contest* $v \in \mathcal{V}$, *the expected equilibrium effort is*

$$\mathbb{E}[X] = \sum_{m=1}^{N} \alpha_m v_m,$$

where

$$\alpha_m = \frac{1}{N+1}\left[\frac{1}{\theta_K} - \sum_{k=1}^{K-1}\left[H_{\geq m}^{N+1}(P_k) + (N-m)H_m^{N+1}(P_k)\right]\left(\frac{1}{\theta_{k+1}} - \frac{1}{\theta_k}\right)\right]. \tag{5}$$

From Lemma 2, the effect of increasing competition by shifting value from a lower-ranked prize m' to a better-ranked prize m is

$$\frac{\partial \mathbb{E}[X]}{\partial v_m} - \frac{\partial \mathbb{E}[X]}{\partial v_{m'}} = \alpha_m - \alpha_{m'}.$$

Hence, the effect is independent of the specific contest $v \in \mathcal{V}$. It depends only on the contest environment $(N+1, \Theta, p)$ and the prizes involved m, m', and can be computed explicitly using Equation (5).

It turns out that for any contest environment $(N+1, \Theta, p)$, shifting value to the best prize $m = N$ always encourages effort. As a consequence, for the design problem, allocating the entire budget to the best prize maximizes expected effort.

Proposition 1 (WTA is optimal)**.** *Fix any contest environment* $(N+1, \Theta, p)$ *with* $|\Theta| > 1$. *For any prize* $m' \in \{1, \ldots, N-1\}$,

$$\alpha_N - \alpha_{m'} > 0.$$

Consequently, among all contests $v \in \mathcal{V}$ *satisfying* $\sum_{m=0}^{N} v_m \leq V$, *the winner-takes-all contest* $v = (0, 0, \ldots, 0, V)$ *uniquely maximizes expected equilibrium effort.*

Proof. From Equation (5), we have that for any prize $m' \in \{1, \ldots, N-1\}$,

$$\alpha_N - \alpha_{m'} = \frac{1}{N+1}\left[\sum_{k=1}^{K-1}\left[H_{\geq m'}^{N+1}(P_k) - H_{\geq N}^{N+1}(P_k) + (N-m')H_{m'}^{N+1}(P_k)\right]\left(\frac{1}{\theta_{k+1}} - \frac{1}{\theta_k}\right)\right].$$

Since $H_{\geq m'}^{N+1}(t) - H_{\geq N}^{N+1}(t) > 0$ for $t \in (0,1)$, and $\frac{1}{\theta_{k+1}} - \frac{1}{\theta_k} > 0$, every term in the sum is strictly positive, so $\alpha_N - \alpha_{m'} > 0$. It follows that for any contest $v \in \mathcal{V}$, transferring value from any lower-ranked prize m' to the top-prize N strictly increases expected effort. □

This result resolves Sisak (2009)'s conjecture in the negative for the case of identical distributions. To contrast with the complete information case, notice that when $|\Theta| = K = 1$, we have from Equation (5) that $\alpha_1 = \alpha_2 = \cdots = \alpha_N = \frac{1}{(N+1)\theta_1}$. Thus, any allocation of the budget across these prizes induces the same expected effort (Barut and Kovenock, 1998). When $|\Theta| > 1$, however, the most efficient types are the cheapest to incentivize, and shifting value to the top prize has an encouraging effect on efficient types that more than offsets the discouraging effect on less efficient types. This extends the optimality result of Moldovanu and Sela (2001), established for the continuum case, to environments with finitely many types. Thus, even a small amount of uncertainty (i.e., incomplete information) is sufficient to make the (most competitive) winner-takes-all contest strictly optimal.

Interestingly, however, while shifting value to the best-ranked prize always encourages effort, shifting value to a better-ranked intermediate prize may not. The qualitative effect of shifting value across intermediate prizes depends on the contest environment. We illustrate when the effect may be negative in the context of a contest environment with binary types.

Proposition 2 (Interior discouragement effect)**.** *Fix any contest environment* $(N+1, \Theta, p)$ *with* $|\Theta| = 2$*. For any prize* $m \in \{2, \dots, N\}$*,*

$$P_1 > \frac{m}{N} \implies \text{ for all } m' < m,\ \alpha_m - \alpha_{m'} < 0.$$

Proof. From Equation (5), we have that for any prize $m \in \{2, \dots, N\}$,

$$\begin{aligned}
\alpha_m - \alpha_{m-1} &= \frac{1}{N+1}\left[\left[H_{m-1}^{N+1}(P_1) + (N-(m-1))H_{m-1}^{N+1}(P_1) - (N-m)H_m^{N+1}(P_1)\right]\left(\frac{1}{\theta_2} - \frac{1}{\theta_1}\right)\right] \\
&= \frac{H_{m-1}^{N+1}(P_1)}{N+1}\left(\frac{1}{\theta_2} - \frac{1}{\theta_1}\right)\left[N-m+2-(N-m)\frac{N-m+2}{m}\frac{P_1}{1-P_1}\right] \\
&= \frac{(N-m+2)H_{m-1}^{N+1}(P_1)}{N+1}\left(\frac{1}{\theta_2} - \frac{1}{\theta_1}\right)\left[1 - \frac{(N-m)}{m}\frac{P_1}{1-P_1}\right].
\end{aligned}$$

It is straightforward to verify that if $P_1 > \frac{m}{N}$, then $\alpha_m - \alpha_{m-1} < 0$.

Now for any $m' < m$,

$$\alpha_m - \alpha_{m'} = (\alpha_m - \alpha_{m-1}) + (\alpha_{m-1} - \alpha_{m-2}) + \cdots + (\alpha_{m'+1} - \alpha_{m'}),$$

and since $P_1 > \frac{m}{N} \geq \frac{j}{N}$ for all $j \in \{m'+1, \dots, m\}$, each consecutive difference is strictly negative, and the result follows.

$\square$

Proposition 2 identifies a sufficient condition under which shifting value to a better-ranked intermediate prize discourages effort. Intuitively, as with shifting value to the best prize, such a transformation discourages inefficient types. However, in contrast to the best prize, the encouraging effect on efficient types is dampened, because the incentive for securing the best prize is also diluted. In fact, the equilibrium effort of the most efficient types may even decline following the transformation, as illustrated in Figure 2. Because this encouraging force is weaker, it may fail to offset the discouragement among inefficient types. In particular, when the inefficient type is sufficiently likely, the overall effect of shifting value to better-ranked intermediate prizes is negative. In the extreme case where $P_1 > \frac{N-1}{N}$, any shift in value from a lower-ranked prize to a better-ranked prize discourages effort, except when the better-ranked prize is the best prize.

The interior discouragement effect may have implications for the design problem in environments where the designer is constrained. For instance, the top prize may be capped by fairness norms or institutional rules, in which case the design problem is precisely one of allocating the left-over budget across intermediate prizes, where the effect binds. Depending upon the contest environment, the designer may prefer to allocate as much as feasible to the best intermediate prizes, or it may prefer to allocate it equally across all intermediate prizes. Beyond the design problem, contests as an institution may arise endogenously, and our results shed light on how changes to such institutions might affect the effort of participating agents. In Section 6, we experimentally test the two main findings of this section: the optimality of the winner-takes-all contest and the interior discouragement effect.

5 General costs

So far, we have focused on contests with linear effort costs. In this section, we extend the analysis to general cost functions. Our approach is to identify conditions under which the effect of competition under linear costs extends to equilibrium under general costs.

Model

Formally, we enrich the model as follows. Fix a contest environment $(N+1, \Theta, p)$ and contest $v \in \mathcal{V}$. If an agent of type $\theta \in \Theta$ wins prize v_m after exerting effort $x \geq 0$, their vNM utility is

$$v_m - \theta c(x),$$

where $c : \mathbb{R}_+ \to \mathbb{R}_+$ is a cost function satisfying $c(0) = 0$, $c'(x) > 0$ for $x > 0$, and $\lim_{x\to\infty} c(x) = \infty$. We maintain these properties throughout and refer to any such c simply as a *cost function.*

We denote a symmetric equilibrium by $(X_1^*, X_2^*, \ldots, X_K^*)$, and let $X^* \sim F^*$ denote the ex-ante equilibrium effort of an arbitrary agent. As before, we let

$$X_v^* : [0,1] \to \mathbb{R}_+$$

denote the quantile function of the ex-ante effort distribution F^*, assigning to each quantile level $t \in [0,1]$ the corresponding equilibrium effort.

Equilibrium

For this Bayesian game, we can equivalently describe agents' choices in terms of the cost of effort $c(x)$ rather than effort itself. In other words, we can reinterpret the game as one in which agents directly choose effort costs, are ranked based on their effort costs (which coincides with their ranking based on effort), and are awarded the corresponding prizes. From this change-of-variable interpretation, it follows that the equilibrium effort under linear costs (Theorem 1) more generally characterizes equilibrium effort costs in the present environment. This leads immediately to the following characterization of equilibrium effort in this more general environment.

Theorem 2 (Equilibrium characterization – general costs)**.** *For any contest environment $(N+1, \Theta, p)$, cost function $c(\cdot)$, and contest $v \in \mathcal{V}$, there exists a unique symmetric equilibrium $(X_1^*, \ldots, X_K^*)$. It is characterized by the quantile function $X_v^* : [0,1] \to \mathbb{R}_+$ of the ex-ante effort distribution: for each $t \in [0,1]$,*

$$X_v^*(t) = c^{-1}(X_v(t)),$$

where $X_v(t)$ is the equilibrium quantile function under linear costs $c(x) = x$ (Theorem 1).

Competition

By the probability integral transform, the quantile level $t = F^*(X^*)$ is ex-ante uniformly distributed on $[0,1]$, and so the expected equilibrium effort is

$$\begin{aligned}\mathbb{E}[X^*] &= \int_0^1 X_v^*(t)\,dt,\\ &= \int_0^1 g(X_v(t))\,dt, \quad \text{where } g = c^{-1}.\end{aligned}$$

The effect of increasing competition by shifting value from a lower-ranked prize m' to a better-ranked prize m is then

$$\frac{\partial \mathbb{E}[X^*]}{\partial v_m} - \frac{\partial \mathbb{E}[X^*]}{\partial v_{m'}} = \int_0^1 g'(X_v(t)) \underbrace{\left(\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}\right)}_{\substack{\text{i) effect on effort if } c(x)=x \\ \text{ii) effect on effort cost } c(X^*)}} dt. \tag{6}$$

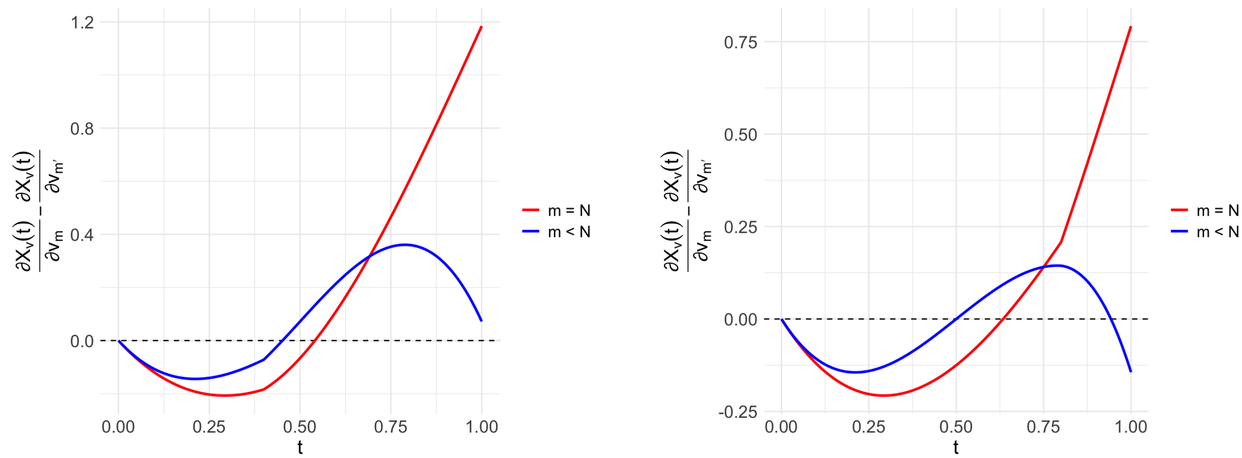
From Equations (3) and (4), we have that

$$\frac{\partial X_v(t)}{\partial v_m} = \frac{H_m^N(t)}{\theta_{k(t)}} - \left[\sum_{j=1}^{k(t)-1} H_m^N(P_j)\left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j}\right)\right]. \tag{7}$$

Equation (6) provides an interpretable and analytically convenient representation for evaluating the effect of shifting value from prize m' to prize m on expected effort. The term $\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}$ represents the effect of the transformation on equilibrium effort at quantile level t under linear costs—equivalently, on equilibrium effort *cost* under cost function $c(\cdot)$. From Equation (7), this effect is independent of both the cost function $c(\cdot)$ and the contest $v \in \mathcal{V}$, and depends only on the contest environment $(N+1, \Theta, p)$ and the specific pair of prizes m, m'. The term $g'(X_v(t))$ captures any influence that the cost function $c(\cdot)$ or the contest $v \in \mathcal{V}$ may have on how the transformation affects effort. It essentially translates the effect on effort cost into the effect on effort at quantile level t, and for our purposes, will be interpreted as simply assigning different weights to different quantile levels. Taking the uniform expectation across quantile levels gives the aggregate effect.

Using the representation in Equation (6), we identify conditions under which the aggregate effect of the transformation under linear costs persists qualitatively under general costs. Notice that the aggregate effect under linear costs is exactly

$$\int_0^1 \left(\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}\right) dt = \alpha_m - \alpha_{m'},$$



(a) Single-crossing ($p = (0.4, 0.6)$) (b) Double-crossing for $m < N$ ($p = (0.8, 0.2)$)

Figure 2: Effect of competition on effort under linear cost ($N = 3, \Theta = \{2, 1\}$ and $m' = 1$)

where α_m for $m \in \{1, \dots, N\}$ are as in Equation (5). It is natural to suspect that in some cases, the sign of $\alpha_m - \alpha_{m'}$ may be preserved under general costs. As discussed earlier, shifting value to better prizes discourages effort among inefficient types and encourages effort among efficient types. However, this encouraging effect is dampened when the better-ranked prize is interior ($m < N$), and may actually even discourage effort among the most efficient types, as illustrated in Figure 2. We show that if, under linear costs, the effect on effort across quantile levels is single-crossing, then positive effects extend to equilibrium under concave costs, while negative effects extend to equilibrium under convex costs.

To illustrate the idea, suppose that the effect across quantile levels under linear costs is single-crossing and that the aggregate effect is non-negative, $\alpha_m - \alpha_{m'} \geq 0$. If the cost function $c(\cdot)$ is concave, then $g(\cdot)$ is convex, so that $g'(X_v(t))$ is increasing in t. Consequently, this term assigns relatively lower weights to smaller quantile levels and larger weights to higher quantile levels. Since the aggregate effect under linear costs is non-negative, it follows that $\frac{\partial \mathbb{E}[X^*]}{\partial v_m} - \frac{\partial \mathbb{E}[X^*]}{\partial v_{m'}} \geq 0$.

We are now ready to state our main result of this section. It identifies conditions under which the effect on effort across quantile levels under linear costs is single-crossing, and shows that, under these conditions, positive effects under linear costs persist under concave costs, while negative effects persist under convex costs.

Theorem 3 (Linear to general)**.** *Fix any contest environment* $(N+1, \Theta, p)$*. Let* m, m' *with* $m > m'$ *be such that either*

(i) $m = N$ *or*

(ii) $\sum_{j=1}^{K-1} \left[H_{m'}^N(P_j) - H_m^N(P_j)\right] \left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j}\right) \geq 0.$

The following statements hold:

(a) If $\alpha_m - \alpha_{m'} \geq 0$*, then for any concave cost* c *and contest* $v \in \mathcal{V}$*,* $\dfrac{\partial \mathbb{E}[X^*]}{\partial v_m} - \dfrac{\partial \mathbb{E}[X^*]}{\partial v_{m'}} \geq 0.$

(b) If $\alpha_m - \alpha_{m'} \leq 0$*, then for any convex cost* c *and contest* $v \in \mathcal{V}$*,* $\dfrac{\partial \mathbb{E}[X^*]}{\partial v_m} - \dfrac{\partial \mathbb{E}[X^*]}{\partial v_{m'}} \leq 0.$

Theorem 3 nests and extends the result of Fang, Noe, and Strack (2020) from complete information to arbitrary contest environments with finite types. To see this, consider the special case of a complete information environment ($|\Theta| = K = 1$). In this case, for any m, m' with $m > m'$, condition (ii) of Theorem 3 is satisfied, and from Lemma 2, $\alpha_m - \alpha_{m'} = 0$. Applying Theorem 3, we get that increasing competition always encourages effort under concave costs and always discourages effort under convex costs, recovering the result of Fang, Noe, and Strack (2020) on the effect of competition under complete information.

Other objectives

While we have focused on expected effort, Theorem 3 can be readily generalized to other quantities of interest. To illustrate, consider the expected maximum effort, another commonly studied objective in contest design (Wasser and Zhang, 2023). From the equilibrium characterization in Theorem 2, the expected maximum effort can be expressed as

$$\mathbb{E}[X^*_{\max}] = (N+1) \int_0^1 g(X_v(t)) t^N dt.$$

Thus, the equilibrium effort of an agent at quantile level t is weighted by the probability that it is the maximum, which is t^N.

The effect of shifting value from a lower-ranked prize m' to a better-ranked prize m on expected maximum effort is then

$$\frac{\partial \mathbb{E}[X^*_{\max}]}{\partial v_m} - \frac{\partial \mathbb{E}[X^*_{\max}]}{\partial v_{m'}} = (N+1)\int_0^1 \underbrace{g'(X_v(t))\, t^N}_{\text{weights}} \underbrace{\left(\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}\right)}_{\text{effect on effort if } c(x)=x} dt. \tag{8}$$

Under conditions (i) or (ii) of Theorem 3, the second term is single-crossing, being negative at small quantile levels and positive at large quantile levels. By exactly the same argument as in the proof of Theorem 3, the effect on expected effort under linear costs then extends qualitatively to expected maximum effort under monotonic weights: positive effects extend when $g'(X_v(t))\, t^N$ is increasing in t, and negative effects extend when it is decreasing. This applies, in particular, when the cost $c(\cdot)$ is concave, as g' and the weights are then increasing in t. As a result, if the contest environment $(N+1, \Theta, p)$ is such that $\alpha_m - \alpha_{m'} > 0$, then for any concave cost $c(\cdot)$ and any contest $v \in \mathcal{V}$, shifting value from m' to m increases expected maximum effort. Note, however, that under convex costs the weights need not be monotone, as g' is then decreasing while t^N is increasing. More broadly, Theorem 3 provides a simple recipe for assessing the effect of increasing competition on various equilibrium objects under arbitrary costs, through the effect it has on expected equilibrium effort under linear costs.

Design problem

Using Theorem 3, we can extend the optimality of winner-takes-all under linear costs, established earlier in Proposition 1, to arbitrary concave costs. Intuitively, the encouraging effect of shifting value from any lower-ranked prize m' to the top prize $m = N$ on the effort of the most efficient types is even stronger under concave costs, as marginal costs are lower at higher effort levels. Formally, such a transformation satisfies condition (i) of Theorem 3, and we know from Proposition 1 that $\alpha_N - \alpha_{m'} > 0$. A straightforward application of Theorem 3 reveals that the transformation encourages effort, so that winner-takes-all is optimal for maximizing expected effort. In fact, the same idea applies to the objective of maximum effort, and more generally to total effort across the top q agents: the probability of ranking among the top q is increasing in the quantile level, so the induced weights remain monotone, and these objectives place even greater weight on the effort of the more efficient types—precisely those that winner-takes-all incentivizes most.

Theorem 4 (WTA is optimal – concave costs)**.** *Fix any contest environment* $(N+1, \Theta, p)$ *and a cost function* $c(\cdot)$ *that is weakly concave. For any* $q \in \{1, \ldots, N+1\}$*, among all contests* $v \in \mathcal{V}$ *satisfying* $\sum_{m=0}^{N} v_m \le V$*, the winner-takes-all contest* $v = (0, 0, \ldots, 0, V)$ *maximizes the expected total equilibrium effort across the top* q *agents.*

Theorem 4 extends the optimality of winner-takes-all under linear and concave costs—established by Moldovanu and Sela (2001) for total effort in the continuum type-space environment—to finite type-spaces and to a broad class of objectives.

With convex costs, however, marginal effort costs are increasing, and so inducing additional effort at higher effort levels becomes progressively more expensive. As a result, concentrating prize value at the top no longer has an unambiguously dominant effect, since the designer must now weigh stronger incentives for efficient types against the rising marginal costs of effort. Nevertheless, for a parametric family of cost functions, we show that the optimality of the winner-takes-all contest persists under moderate convexity. In other words, as long as costs are not too convex, the incentive gains from concentrating prize value at the top continue to outweigh the associated marginal cost increases. The following proposition formalizes this.

Proposition 3 (WTA is optimal – moderately convex costs)**.** *Fix any contest environment* $(N+1, \Theta, p)$ *with* $|\Theta| > 1$ *and consider cost function* $c(x) = x^{\alpha}$*. For any* $q \in \{1, \ldots, N+1\}$*, there exists* $\alpha^* > 1$ *such that if* $\alpha \in [1, \alpha^*)$*, then among all contests* $v \in \mathcal{V}$ *satisfying* $\sum_{m=0}^{N} v_m \leq V$*, the winner-takes-all contest* $v = (0, 0, \ldots, 0, V)$ *maximizes the expected total equilibrium effort across the top* q *agents.*

Together, our results establish the winner-takes-all contest as being robustly optimal: it maximizes the total effort across the top q agents provided effort costs are not too convex.

In particular, Proposition 3 reveals a sharp contrast with the complete-information environment, where the winner-takes-all contest instead minimizes total effort under convex costs (Fang, Noe, and Strack, 2020). Our analysis suggests a reconciliation of these contrasting findings in that the supremacy of the winner-takes-all contest over alternative prize structures in incomplete-information environments diminishes as the environment approaches complete information. This pattern is illustrated in Figure 3, which shows that in a binary-type environment, the effect of shifting value to the top prize—while always positive—shrinks as the probability of the inefficient type, p_1, approaches 0 or 1. Based on these findings, we conjecture that the degree of convexity required to overturn the optimality of the winner-takes-all contest likewise decreases as the environment approaches complete information.

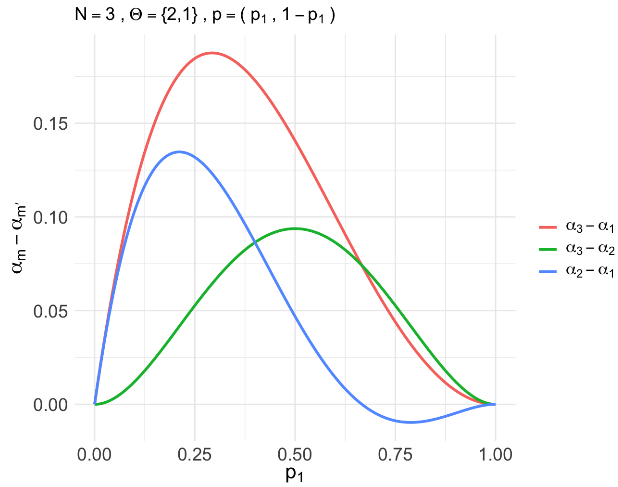


Figure 3: Effect of increasing competition: Complete and Incomplete information.

6 Experiment

In this section, we present findings from an incentivized experiment designed to test the equilibrium predictions of our model regarding the effect of competition on effort. Our primary objective is to test the optimality of the (most competitive) winner-takes-all contest under linear costs, originally established by Moldovanu and Sela (2001) in a continuum type-space setting. Our result for the finite type-space model establishes its robustness under incomplete information and enables experimental investigation. We test this by comparing effort levels in a winner-takes-all structure with those in less competitive alternatives.

Our second goal is to test for the interior discouragement effect of competition, which our model predicts emerges when inefficient types are sufficiently likely. To test this, we consider an environment where this condition holds and gradually increase the competitiveness of the prize structure within an interior range, where the model predicts a decrease in expected equilibrium effort. Below, we present the experimental design, implementation details, and our findings.

6.1 Experimental Design

In our experiment, subjects competed in groups of four for monetary prizes. Effort levels were chosen independently and privately, without any possibility of communication. There were two possible types (marginal costs of effort): 2 and 1, assigned with probabilities 80%

and 20%, respectively.[7] We implemented the strategy method. Each subject participated in all four contests, submitting two effort choices per contest—one for each possible type—for a total of eight decisions per subject. At the end of the experiment, each subject was independently assigned a type according to the type distribution, and the corresponding decisions were used to determine payoffs.

The experimental treatments varied the contest $v = (v_0, v_1, v_2, v_3)$, which awards prizes based on effort rankings: v_3 for the highest effort, v_2 for the second highest, v_1 for the third highest, and $v_0 = 0$ for the lowest. The most competitive treatment, *WTA*, is a winner-takes-all structure with $(0, 0, 0, 100)$. We consider three other treatments with the same total prize but progressively less competitive: *High* with $v = (0, 0, 25, 75)$, *Med* with $v = (0, 0, 50, 50)$, and *Low* with $v = (0, 25, 25, 50)$. Table 1 summarizes the treatments and their equilibrium effort predictions.

The contest order was randomized across subjects. The group composition in each contest was also randomized and the subjects were unable to identify with each other between contests. Feedback on contest outcomes was withheld until the end of the experiment, however on-screen information informed subjects of their potential earnings for each possible prize they could win (i.e. prize minus effort costs) to ensure proper comprehension of the payoffs. Subjects were paid for one randomly selected contest, with tokens converted to U.S. dollars at a rate of 50:1, plus a $2 show up fee. The protocols for subject group matching, feedback, and payment were chosen to minimize reputation-building or repeated-play concerns, thereby inducing the one-shot nature of the game under study.

We recruited 700 subjects from Prolific, an online labor market, during April 2025. We administered a comprehension quiz to ensure that subjects understood the instructions, the structure of the contests, and how the earnings were determined. Subjects who failed to pass the quiz after a second attempt were not allowed to continue, resulting in a final sample of 445 subjects.[8]

[7] In terms of our notation, the contest environment $(N + 1, \Theta, p)$ is such that $N + 1 = 4$, the type-space $\Theta = \{2, 1\}$, and distribution $p = (0.8, 0.2)$.

[8] The experimental interface and instructions are provided here.

Treatment	$(0, v_1, v_2, v_3)$	Equilibrium Effort ($\mathbb{E}[X]$)			Observed Effort		
		$\theta = 2$	$\theta = 1$	Pooled	$\theta = 2$	$\theta = 1$	Pooled
WTA	$(0, 0, 0, 100)$	6.4	48.2	14.76	40.5	52.8	42.96
High	$(0, 0, 25, 75)$	8.0	37.0	13.80	37.4	46.9	39.30
Med	$(0, 0, 50, 50)$	9.6	25.8	12.84	36.5	43.3	37.86
Low	$(0, 25, 25, 50)$	10.2	24.6	13.08	35.9	42.9	37.30

Table 1: Equilibrium and Observed Efforts, by Treatment and Cost Type

Note: $\theta = 2$ is the inefficient (high-cost) type, drawn with probability 0.8; $\theta = 1$ is the efficient (low-cost) type, drawn with probability 0.2. Pooled values are the corresponding probability-weighted averages.

6.2 Experimental Results

Table 1 summarizes our findings on mean effort. The mean effort is highest under the *WTA*. Also, efforts are higher for the efficient type ($\theta = 1$) and lower for the inefficient type ($\theta = 2$), as expected. Regression analysis, reported in column 1 of Table 4 in Appendix E, confirms the statistical significance of these findings.

To directly test whether behavior aligns with the theory, we use Lemma 2, which represents the expected equilibrium effort as a linear combination of the contest prizes. For the environment considered in the experiment, we have that for any contest $v \in \mathcal{V}$,

$$\mathbb{E}[X] = 0.119 \cdot v_1 + 0.109 \cdot v_2 + 0.148 \cdot v_3 \,.$$

In this representation, observe that the largest coefficient is on the best prize (v_3), which implies that shifting value from any lower-ranked prize to the best prize increases expected effort, thus resulting in the optimality of the WTA prize structure. The interior discouragement effect is reflected in the fact that the coefficient v_2 is smaller than that of v_1, so that increasing competition by shifting value from lower-ranked prize v_1 to higher-ranked prize v_2 decreases expected equilibrium effort.

We can also decompose the expected effort by type and obtain that

$$\mathbb{E}[X \mid \theta = 2] = 0.152 \cdot v_1 + 0.128 \cdot v_2 + 0.064 \cdot v_3$$

and

$$\mathbb{E}[X \mid \theta = 1] = -0.014 \cdot v_1 + 0.034 \cdot v_2 + 0.482 \cdot v_3.$$

Observe that the best prize (v_3) has the largest coefficient for the efficient type and the smallest for the inefficient type. This suggests that the overall optimality of the WTA structure is primarily driven by the strong incentive it creates for the efficient type. For the interior prizes, shifting value from v_1 to v_2 encourages effort from the efficient type, but discourages effort from the inefficient type. The overall interior discouragement effect arises because the inefficient type is relatively more likely.

Table 2 presents results from a linear regression of effort choices on the three prizes.[9] We find that, in contrast to theoretical predictions, the best prize carries the largest coefficient not only for the efficient type, but also for the inefficient type. Moreover, both the intermediate prizes appear to be overweighted in observed effort choices relative to theoretical expectations, for both types. Even so, in the aggregate, the best prize has the largest coefficient for mean effort, and therefore, shifting value from any lower-ranked prize to the best prize is empirically correlated with an increase in effort. This finding speaks to the global optimality of the winner-takes-all contest, in line with the theory.

For the interior discouragement effect, we find that the difference in the estimated coefficients is negligible (0.006), and statistically insignificant (the Wald test for the coefficients being equal yields a p-value of 0.850). Although we do not find a significant negative effect of shifting value from v_1 to v_2, we also do not find a positive effect. We interpret this result as partially in line with equilibrium behavior, as it provides evidence that increasing the competitiveness of the prize structure need not lead to higher effort.

6.3 Discussion of Experimental Results

In the preceding subsection, we focused on testing the theoretical predictions. While we find qualitative support for the main comparative statics of the model, our results also reveal evidence of rent dissipation in the form of excessive effort, especially from the inefficient type. This phenomenon of over-provision has been documented throughout the contest literature, though it is known to attenuate with experience (Lugovskyy, Puzzello, and Tucker, 2010).

[9] Since the expected equilibrium effort is zero when all prizes are zero, we estimate the regression without a constant to align directly with the theoretical specification.

Prize	Equilibrium Weight			Estimated Coefficient		
	$\theta = 2$	$\theta = 1$	Pooled	$\theta = 2$	$\theta = 1$	Pooled
First place prize (v_3)	0.064	0.482	0.148	0.401	0.524	0.426
Second place prize (v_2)	0.128	0.034	0.109	0.321	0.335	0.324
Third place prize (v_1)	0.152	−0.014	0.119	0.314	0.334	0.318

Table 2: Linear Decomposition of the Expected Effort: Equilibrium and Regression Results
Note: Lemma 2 establishes that equilibrium effort is a linear function of the contest prize vector. The columns labeled "Equilibrium Weight" report the theoretical coefficients for each prize by type and in the aggregate; "Estimated Coefficient" columns report the corresponding regression estimates. $\theta = 2$ is the inefficient type and $\theta = 1$ the efficient type. Full regression results are in Table 3.

In related experiments on all-pay auctions with private valuations (Barut, Kovenock, and Noussair, 2002; Noussair and Silver, 2006) and contests (Müller and Schotter, 2010), the over-bidding is largely attributed to the efficient types, whereas in our data the inefficient type substantially overbids and accounts for most of the aggregate over-provision. A plausible reason might be that the inefficient type is drawn with probability 0.8, so most subjects expect to face opponents of their own type, which may make competing appear worthwhile even when equilibrium prescribes low effort.

We briefly discuss the non-strategic determinants of behavior by examining how individual characteristics correlate with effort choices, as is common in the analysis of experimental data. For this purpose, we elicited self-reported measures of risk attitudes (willingness to take risks on a 0–10 Likert scale) and competitiveness (willingness to compete on a 0–10 Likert scale) at the end of the experiment to proxy for the intrinsic *joy of winning* that subjects may derive beyond the pecuniary earnings. We also control for gender. These factors have been shown to influence behavior in prior studies on contests (Dechenaux, Kovenock, and Sheremeta, 2015). Regression results, reported in column 2 of Table 4 in Appendix E, show that subjects who declare a higher willingness to take risks (coefficient 1.528, $p < 0.01$) and claim to be more competitive (coefficient 1.063, $p < 0.05$) choose significantly higher efforts. We find no significant gender differences in effort choices.

7 Conclusion

We study all-pay contests with finite type-spaces and show that the most competitive winner-takes-all contest maximizes the total effort of the top q agents under linear, concave, and moderately convex costs. At the same time, the effect of competition is not monotonic: we uncover an interior discouragement effect, whereby shifting value to better-ranked intermediate prizes may discourage effort when inefficient types are relatively likely. We complement the theory with an experiment, which reveals significant over-provision of effort, especially among inefficient types, though aggregate patterns broadly align with the model's comparative statics. Our analysis rests on a novel quantile representation of equilibrium effort, which extends the assortative allocation ideas typically used in continuum type-space environments to finite type-space environments, enabling tractable analysis of otherwise complex mixed-strategy equilibria.

Our findings suggest several promising avenues for future research. First, our results point toward a reconciliation of the contrasting findings in complete- and incomplete-information environments: the degree of cost convexity required to overturn the optimality of winner-takes-all appears to shrink as the environment approaches complete information. Investigating this formally is an important next step. Second, exploring non-parametric type-spaces, especially unordered ones, may yield new insights into contest design. More broadly, the techniques we develop may apply to a wider class of contest and auction environments, and may prove especially valuable in settings where mixed-strategy equilibria have previously impeded tractable analysis. Finally, finite type-spaces offer a natural framework for experimental work: with explicit equilibrium characterization and convergence results, our analysis provides a strong foundation for testing the many theoretical predictions in the literature on contests under incomplete information.

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A Proofs for Section 3 (Equilibrium)

Lemma 1 (Equilibrium structure)**.** *For any contest environment* $(N+1, \Theta, p)$ *and contest* $v \in \mathcal{V}$*, if* $(X_1, \ldots, X_K)$ *is an equilibrium, then there exist boundary points*

$$0 = b_0 < b_1 < \cdots < b_K$$

such that, for each k*,* X_k *is continuously distributed on* $[b_{k-1}, b_k]$*.*

Proof. Suppose $(X_1, \ldots, X_K)$ is an equilibrium, and $X \sim F$ is the ex-ante equilibrium effort. We establish the result in three steps.

Step 1: For each k, X_k is a continuous random variable.

Suppose, for contradiction, that $\Pr[X_k = x] > 0$ for some x. Under the given profile, there is a positive probability that all $N+1$ agents choose effort x, in which case ties are broken uniformly at random. If a type-θ_k agent deviates to $x+\epsilon$, it obtains a discontinuous jump in expected prize (since $v_N > v_0$) while the additional cost $\theta_k(x+\epsilon) - \theta_k x$ can be made arbitrarily small. Hence the deviation is profitable, a contradiction.

Step 2: There exists $b_K > 0$ such that X is continuously distributed on $[0, b_K]$.

Suppose there exists an interval (d_1, d_2) that is not contained in the support of X. Consider a type whose support contains d_2. By deviating to d_1, the agent obtains the same expected prize while incurring strictly lower cost, yielding a profitable deviation. Hence the support of X cannot contain gaps. Moreover, the support must include 0: if the minimum of the support were some $a > 0$, an agent exerting effort a outperforms the other agents with probability 0, and so can deviate to 0, preserving the expected prize while saving on cost. Finally, the support is bounded above, since any effort $x > \frac{v_N}{\theta_K}$ yields a strictly negative payoff and is thus dominated by zero effort. It follows that the support is a convex and compact subset of $\mathbb{R}_+$ containing 0, i.e., an interval $[0, b_K]$ for some $b_K > 0$.

Step 3: There exist $b_1 < b_2 < \cdots < b_K$ such that the support of X_k is $[b_{k-1}, b_k]$.

Let $x < y$ lie in the support of X_k. Since a type-θ_k agent must be indifferent between the two,

$$\pi_v(F(y)) - \theta_k y = \pi_v(F(x)) - \theta_k x.$$

For a type-θ_j agent, the payoff difference between choosing y and x is

$$\pi_v(F(y)) - \theta_j y - \big(\pi_v(F(x)) - \theta_j x\big) = (\theta_k - \theta_j)(y - x).$$

This expression is strictly positive if $\theta_j < \theta_k$ and strictly negative if $\theta_j > \theta_k$. Hence no other type is indifferent between x and y, implying that the supports of X_j and X_k intersect in at most boundary points. The ordering follows from the observation that if $\theta_j < \theta_k$, a type-θ_j agent obtains a higher payoff from y than x. Therefore there exist boundary points $0 = b_0 < b_1 < \cdots < b_K$ such that the support of X_k is $[b_{k-1}, b_k]$. □

Theorem 1 (Equilibrium characterization)**.** *For any contest environment* $(N+1, \Theta, p)$ *and contest* $v \in \mathcal{V}$*, there exists a unique symmetric equilibrium* $(X_1, \ldots, X_K)$*. It is characterized by the quantile function* $X_v : [0,1] \to \mathbb{R}_+$ *of the ex-ante effort distribution* F*: for each* $t \in [0,1]$*,*

$$X_v(t) = \frac{\pi_v(t) - u_{k(t)}}{\theta_{k(t)}}, \tag{3}$$

where $k(t) = \max\{k : P_{k-1} \leq t\}$*, and the equilibrium utilities are given by*

$$u_k = \theta_k \left[\sum_{j=1}^{k-1} \pi_v(P_j) \left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j} \right) \right]. \tag{4}$$

Proof. Lemma 1 and Equation (2) imply that any symmetric equilibrium must take the stated form, given the equilibrium utilities. We first establish the utilities by induction, and then verify that the resulting profile is indeed an equilibrium.

Base case. Since zero effort lies in the support of the type-θ_1 strategy and yields payoff 0, we have $u_1 = 0$.

Induction step. Suppose

$$u_k = \theta_k \left[\sum_{j=1}^{k-1} \pi_v(P_j) \left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j} \right) \right].$$

Evaluating the equilibrium effort at $t = P_k$ yields

$$\begin{aligned} X_v(P_k) &= \frac{\pi_v(P_k) - u_k}{\theta_k} \\ &= \frac{\pi_v(P_k)}{\theta_k} - \sum_{j=1}^{k-1} \pi_v(P_j) \left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j} \right) \\ &= b_k. \end{aligned}$$

Since b_k lies in the support of the type-θ_{k+1} strategy, their equilibrium utility is

$$\begin{aligned}
u_{k+1} &= \pi_v(P_k) - \theta_{k+1} b_k \\
&= \pi_v(P_k) - \theta_{k+1} \left[\frac{\pi_v(P_k)}{\theta_k} - \sum_{j=1}^{k-1} \pi_v(P_j) \left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j} \right) \right] \\
&= \theta_{k+1} \left[\sum_{j=1}^{k} \pi_v(P_j) \left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j} \right) \right],
\end{aligned}$$

which establishes the inductive claim.

Now to verify that this strategy is indeed an equilibrium, fix any agent and type θ_k, and assume all other N agents are playing this strategy. Since each opponent's ex-ante effort distribution is F, an effort $x \geq 0$ outperforms an arbitrary opponent with probability $F(x)$, and thus yields the payoff

$$u(x) = \pi_v(F(x)) - \theta_k x.$$

By construction, $u(x) = u_k$ for all $x \in [b_{k-1}, b_k]$; we show that $u(x) \leq u_k$ for all $x \notin [b_{k-1}, b_k]$. Consider $x \in [b_{j-1}, b_j]$ for some $j \neq k$. Since a type-θ_j agent is indifferent between x and any point of $[b_{j-1}, b_j]$,

$$\pi_v(F(x)) - \theta_j x = \pi_v(F(b_j)) - \theta_j b_j,$$

and therefore

$$u(x) = u(b_j) - (\theta_j - \theta_k)(b_j - x).$$

If $j < k$, then $\theta_j > \theta_k$ and $b_j \geq x$, so $u(x) \leq u(b_j)$: the agent weakly prefers the upper endpoint of any interval below its own. Chaining over $j = 1, \ldots, k-1$ yields $u(x) \leq u(b_{k-1}) = u_k$ for all $x \leq b_{k-1}$. Similarly, writing $u(x) = u(b_{j-1}) - (\theta_j - \theta_k)(b_{j-1} - x)$, if $j > k$ then $\theta_j < \theta_k$ and $b_{j-1} \leq x$, so $u(x) \leq u(b_{j-1})$: the agent weakly prefers the lower endpoint of any interval above its own. Chaining over $j = k+1, \ldots, K$ yields $u(x) \leq u(b_k) = u_k$ for all $x \in (b_k, b_K]$. Finally, for $x > b_K$, we have $u(x) = v_N - \theta_k x < v_N - \theta_k b_K = u(b_K) \leq u_k$, where the last inequality follows from the chaining argument above. Hence no deviation is profitable, and the profile is an equilibrium. $\square$

B Proofs for Section 4 (Competition)

Lemma 2 (Expected effort)**.** *For any contest environment $(N+1, \Theta, p)$ and contest $v \in \mathcal{V}$, the expected equilibrium effort is*

$$\mathbb{E}[X] = \sum_{m=1}^{N} \alpha_m v_m,$$

where

$$\alpha_m = \frac{1}{N+1}\left[\frac{1}{\theta_K} - \sum_{k=1}^{K-1}\left[H_{\geq m}^{N+1}(P_k) + (N-m)H_m^{N+1}(P_k)\right]\left(\frac{1}{\theta_{k+1}} - \frac{1}{\theta_k}\right)\right]. \tag{5}$$

Proof. Since $t = F(X)$ is uniformly distributed on $[0,1]$,

$$\begin{aligned}
\mathbb{E}[X] &= \int_0^1 X_v(t)dt \\
&= \int_0^1 \frac{\pi_v(t) - u_{k(t)}}{\theta_{k(t)}}dt && \text{(Theorem 1)} \\
&= \sum_{k=1}^{K} p_k \cdot \frac{1}{\theta_k} \cdot \left[\int_{P_{k-1}}^{P_k} \frac{\pi_v(t)}{p_k}dt - u_k\right].
\end{aligned}$$

Notice that for any $k \in [K]$, $\int_{P_{k-1}}^{P_k} \frac{\pi_v(t)}{p_k}dt$ is the expected prize awarded to a type-θ_k agent. To compute this, we instead compute the ex-ante expected total prize awarded to type-θ_k agents. Notice that for any prize $m \in \{0, \ldots, N\}$, the ex-ante probability that this prize is awarded to a type-θ_k agent is simply

$$\left[H_{\geq m+1}^{N+1}(P_k) - H_{\geq m+1}^{N+1}(P_{k-1})\right].$$

Thus, the ex-ante expected total prize awarded to type-θ_k agents is

$$\sum_{m=1}^{N} v_m \left[H_{\geq m+1}^{N+1}(P_k) - H_{\geq m+1}^{N+1}(P_{k-1})\right].$$

By an alternative calculation, which entails adding up over the $N+1$ agents, this expectation should equal

$$(N+1) \cdot p_k \cdot \int_{P_{k-1}}^{P_k} \frac{\pi_v(t)}{p_k}dt.$$

Equating these two, we get that

$$\int_{P_{k-1}}^{P_k} \pi_v(t)dt = \frac{\sum_{m=1}^{N} v_m \left[H_{\geq m+1}^{N+1}(P_k) - H_{\geq m+1}^{N+1}(P_{k-1})\right]}{N+1}.$$

Alternatively, we can also directly use the following fact to compute this integral:

$$\frac{\partial H_{\geq m+1}^{N+1}(t)}{\partial t} = (N+1) H_m^N(t)$$

Substituting this in the above representation, we get

$$\mathbb{E}[X] = \sum_{k=1}^{K} \frac{1}{(N+1)\theta_k} \sum_{m=1}^{N} v_m \left[H_{\geq m+1}^{N+1}(P_k) - H_{\geq m+1}^{N+1}(P_{k-1})\right] - \sum_{k=1}^{K} \frac{p_k u_k}{\theta_k}.$$

From here, we write

$$\mathbb{E}[X] = \sum_{m=1}^{N} \alpha_m v_m$$

where

$$\begin{aligned}
\alpha_m &= \sum_{k=1}^{K} \frac{\left[H_{\geq m+1}^{N+1}(P_k) - H_{\geq m+1}^{N+1}(P_{k-1})\right]}{(N+1)\theta_k} - \sum_{k=1}^{K} p_k \sum_{j=1}^{k-1} H_m^N(P_j) \left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j}\right) \\
&= \sum_{k=1}^{K} \frac{\left[H_{\geq m+1}^{N+1}(P_k) - H_{\geq m+1}^{N+1}(P_{k-1})\right]}{(N+1)\theta_k} - \sum_{k=1}^{K-1} (1-P_k) H_m^N(P_k) \left(\frac{1}{\theta_{k+1}} - \frac{1}{\theta_k}\right) \\
&= \frac{1}{N+1}\left[\frac{1}{\theta_K} - \sum_{k=1}^{K-1} H_{\geq m+1}^{N+1}(P_k)\left(\frac{1}{\theta_{k+1}} - \frac{1}{\theta_k}\right)\right] - \frac{(N+1-m)}{N+1} \sum_{k=1}^{K-1} \left[H_m^{N+1}(P_k)\left(\frac{1}{\theta_{k+1}} - \frac{1}{\theta_k}\right)\right] \\
&= \frac{1}{N+1}\left[\frac{1}{\theta_K} - \sum_{k=1}^{K-1} \left[H_{\geq m}^{N+1}(P_k) + (N-m) H_m^{N+1}(P_k)\right]\left(\frac{1}{\theta_{k+1}} - \frac{1}{\theta_k}\right)\right].
\end{aligned}$$

□

C Proofs for Section 5 (General costs)

Lemma 3. *Suppose $a_2 : [0,1] \to \mathbb{R}$ is such that there exists $t^* \in [0,1]$ so that $a_2(t) \leq 0$ for $t \leq t^*$ and $a_2(t) \geq 0$ for $t \geq t^*$. Then, for any increasing function $a_1 : [0,1] \to \mathbb{R}$,*

$$\int_0^1 a_1(t) a_2(t) dt \geq a_1(t^*) \int_0^1 a_2(t) dt.$$

Proof. Observe that

$$\begin{aligned}
\int_0^1 a_1(t) a_2(t) dt &= \int_0^{t^*} a_1(t) a_2(t) dt + \int_{t^*}^1 a_1(t) a_2(t) dt \\
&\geq \int_0^{t^*} a_1(t^*) a_2(t) dt + \int_{t^*}^1 a_1(t^*) a_2(t) dt
\end{aligned}$$

$$= a_1(t^*) \int_0^1 a_2(t)dt.$$

$\square$

Theorem 3 (Linear to general)**.** *Fix any contest environment $(N+1, \Theta, p)$. Let m, m' with $m > m'$ be such that either*

(i) $m = N$ *or*

(ii) $\sum_{j=1}^{K-1} \left[H_{m'}^N(P_j) - H_m^N(P_j)\right] \left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j}\right) \geq 0.$

The following statements hold:

(a) If $\alpha_m - \alpha_{m'} \geq 0$, then for any concave cost c and contest $v \in \mathcal{V}$, $\dfrac{\partial \mathbb{E}[X^*]}{\partial v_m} - \dfrac{\partial \mathbb{E}[X^*]}{\partial v_{m'}} \geq 0.$

(b) If $\alpha_m - \alpha_{m'} \leq 0$, then for any convex cost c and contest $v \in \mathcal{V}$, $\dfrac{\partial \mathbb{E}[X^*]}{\partial v_m} - \dfrac{\partial \mathbb{E}[X^*]}{\partial v_{m'}} \leq 0.$

Proof. From Equation (6), we have that for any contest $v \in \mathcal{V}$ and prizes m, m' with $m > m'$,

$$\frac{\partial \mathbb{E}[X^*]}{\partial v_m} - \frac{\partial \mathbb{E}[X^*]}{\partial v_{m'}} = \int_0^1 g'\left(X_v(t)\right) \left(\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}\right) dt$$

where, from Equation (7), we obtain that

$$\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}} = \left(\frac{H_m^N(t) - H_{m'}^N(t)}{\theta_{k(t)}}\right) - \left[\sum_{j=1}^{k(t)-1} \left(H_m^N(P_j) - H_{m'}^N(P_j)\right) \left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j}\right)\right].$$

From here, one can verify that

1. $\left.\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}\right|_{t=0} = 0$

2. $\left.\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}\right|_{t=1} = \begin{cases} \frac{1}{\theta_K} - \left[\sum_{j=1}^{K-1} \left(H_N^N(P_j) - H_{m'}^N(P_j)\right) \left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j}\right)\right] & \text{if } m = N \\ \left[\sum_{j=1}^{K-1} \left(H_{m'}^N(P_j) - H_m^N(P_j)\right) \left(\frac{1}{\theta_{j+1}} - \frac{1}{\theta_j}\right)\right] & \text{otherwise} \end{cases}$

3. $\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}$ is continuous in t

4. $\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}$ is differentiable at all t, except when $t = P_k$. At any $t \in (0,1)$ such that $t \neq P_k$, the derivative has the same sign as the derivative of $H_m^N(t) - H_{m'}^N(t)$ with respect to t.

The conditions (i) or (ii) ensure that $\left.\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}\right|_{t=1} \geq 0$. Together with the above properties, this implies that $\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}$ is single-crossing: there is some $t^* \in [0,1]$ such that $\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}} \leq 0$ for $t \in [0,t^*]$, and $\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}} \geq 0$ for $t \in [t^*,1]$.

Now for the statement in (a), suppose $\alpha_m - \alpha_{m'} \geq 0$, and consider any concave cost $c(\cdot)$ and contest $v \in \mathcal{V}$. Since c is concave, $g = c^{-1}$ is convex, and thus, $g'(X_v(t))$ is increasing in t. Applying Lemma 3 with $a_1(t) = g'(X_v(t))$ and $a_2(t) = \frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}$ gives

$$\begin{aligned}\frac{\partial \mathbb{E}[X^*]}{\partial v_m} - \frac{\partial \mathbb{E}[X^*]}{\partial v_{m'}} &\geq g'(X_v(t^*)) \int_0^1 \left(\frac{\partial X_v(t)}{\partial v_m} - \frac{\partial X_v(t)}{\partial v_{m'}}\right) dt \\ &= g'(X_v(t^*))\,(\alpha_m - \alpha_{m'})\end{aligned}$$

and the result follows. An analogous argument applies for the case where c is convex.

□

Theorem 4 (WTA is optimal – concave costs)**.** *Fix any contest environment* $(N+1, \Theta, p)$ *and a cost function* $c(\cdot)$ *that is weakly concave. For any* $q \in \{1,\dots,N+1\}$*, among all contests* $v \in \mathcal{V}$ *satisfying* $\sum_{m=0}^{N} v_m \leq V$*, the winner-takes-all contest* $v = (0,0,\dots,0,V)$ *maximizes the expected total equilibrium effort across the top* q *agents.*

Proof. Fix any $q \in \{1,\dots,N+1\}$, and let $X^*_{(q)}$ denote the total equilibrium effort across the top q agents. An agent at quantile level t ranks among the top q agents if and only if it outperforms at least $N+1-q$ agents, which occurs with probability $H^N_{\geq N+1-q}(t)$. Hence the expected total equilibrium effort across the top q agents is

$$\mathbb{E}[X^*_{(q)}] = (N+1) \int_0^1 g(X_v(t))\, H^N_{\geq N+1-q}(t)\, dt.$$

Given any contest $v \in \mathcal{V}$, the effect of shifting value from prize m' to the best prize $m = N$ is then

$$\frac{\partial \mathbb{E}[X^*_{(q)}]}{\partial v_N} - \frac{\partial \mathbb{E}[X^*_{(q)}]}{\partial v_{m'}} = (N+1) \int_0^1 \underbrace{g'(X_v(t))\, H^N_{\geq N+1-q}(t)}_{\text{increasing weights}} \underbrace{\left(\frac{\partial X_v(t)}{\partial v_N} - \frac{\partial X_v(t)}{\partial v_{m'}}\right)}_{\text{single-crossing}} dt.$$

Observe that $H^N_{\geq N+1-q}(t)$ is increasing in t: higher quantile raises the probability of ranking among the top q. Under concave c, g' is increasing, so the weight $g'(X_v(t))H^N_{\geq N+1-q}(t)$ is increasing in t. The second factor is single-crossing, and $\alpha_N - \alpha_{m'} \geq 0$ by Proposition 1.

Applying Lemma 3 with $a_1(t) = g'(X_v(t))H^N_{\geq N+1-q}(t)$ and $a_2(t) = \frac{\partial X_v(t)}{\partial v_N} - \frac{\partial X_v(t)}{\partial v_{m'}}$ yields

$$\frac{\partial \mathbb{E}[X^*_{(q)}]}{\partial v_N} - \frac{\partial \mathbb{E}[X^*_{(q)}]}{\partial v_{m'}} \geq 0.$$

Hence shifting value toward the best prize increases the expected total equilibrium effort across the top q agents, and the winner-takes-all contest is optimal for any q. □

Proposition 3 (WTA is optimal – moderately convex costs)**.** *Fix any contest environment $(N+1, \Theta, p)$ with $|\Theta| > 1$ and consider cost function $c(x) = x^\alpha$. For any $q \in \{1, \ldots, N+1\}$, there exists $\alpha^* > 1$ such that if $\alpha \in [1, \alpha^*)$, then among all contests $v \in \mathcal{V}$ satisfying $\sum_{m=0}^{N} v_m \leq V$, the winner-takes-all contest $v = (0, 0, \ldots, 0, V)$ maximizes the expected total equilibrium effort across the top q agents.*

Proof. Fix any $q \in \{1, \ldots, N+1\}$. Given any contest $v \in \mathcal{V}$, the effect of shifting value from prize m' to the best prize $m = N$ is

$$\frac{\partial \mathbb{E}[X^*_{(q)}]}{\partial v_N} - \frac{\partial \mathbb{E}[X^*_{(q)}]}{\partial v_{m'}} = (N+1) \int_0^1 g'(X_v(t))\, H^N_{\geq N+1-q}(t) \, \left(\frac{\partial X_v(t)}{\partial v_N} - \frac{\partial X_v(t)}{\partial v_{m'}} \right) dt,$$

where

$$g'(X_v(t)) = \frac{1}{\alpha}\, X_v(t)^{\frac{1}{\alpha}-1}.$$

From the proof of Theorem 4, we know that at $\alpha = 1$ this expression is strictly positive. By continuity of the expression in v and the curvature parameter α, there exist $\epsilon_v > 0$ and $\delta_v > 0$ such that for all contests v' in an ϵ_v-neighborhood of v, and all $\alpha < 1 + \delta_v$, the effect of the transformation remains strictly positive.

The collection of such neighborhoods of contests forms an open cover of the feasible set of contests, which is compact. We may therefore extract a finite subcover. Let $\delta_{m'}$ denote the minimum of the corresponding δ_v across this finite subcover. It follows that for all feasible contests v and all $\alpha < 1 + \delta_{m'}$, shifting value from prize m' to the top prize increases expected effort.

Repeating this argument for each interior prize $m' \in \{1, \ldots, N-1\}$ and taking the minimum across these interior prizes yields $\alpha^* > 1$ such that, if $\alpha \in [1, \alpha^*)$, starting from any feasible contest $v \in \mathcal{V}$, shifting value from any lower-ranked prize to the best prize increases total equilibrium effort across the top q agents. □

D Convergence

In this section, we establish an equilibrium convergence result for the continuum type-space. Specifically, we show that if a sequence of (parametric) finite type-space distributions converges to a differentiable distribution over a continuum type-space, then the corresponding sequence of mixed-strategy equilibria converges to the pure-strategy equilibrium in the continuum model. Intuitively, as the finite type-space becomes large, the interval over which a given type mixes shrinks, and essentially converges to the effort level prescribed by the pure-strategy equilibrium under the continuum type-space. Thus, the equilibrium in an appropriate and sufficiently large finite-type space provides a reasonable approximation to the equilibrium strategy under the continuum type-space, and vice versa.

We begin by recalling the symmetric equilibrium under a (parametric) continuum type-space (Moldovanu and Sela, 2001). For this section, we focus on the linear cost case ($c(x) = x$), which is without loss of generality due to the equivalence between convergence in effort cost and in effort.

Lemma 4 (Equilibrium for continuum type-space)**.** *Suppose there are $N+1$ agents, each with a private type (marginal cost of effort) drawn from $\Theta = [\underline{\theta}, \overline{\theta}]$ according to a differentiable CDF $G : [\underline{\theta}, \overline{\theta}] \to [0,1]$. For any contest $v \in \mathcal{V}$, there is a unique symmetric Bayes-Nash equilibrium and it is such that for any $\theta \in \Theta$,*

$$X(\theta) = \int_{\theta}^{\overline{\theta}} \frac{\pi_v'(1-G(t))g(t)}{t} dt.$$

Proof. Suppose N agents are playing a strategy $X : [\underline{\theta}, \overline{\theta}] \to \mathbb{R}_+$ so that if an agent's type is θ, it exerts effort $X(\theta)$. Further suppose that $X(\theta)$ is decreasing in θ. Now we want to find the remaining agent's best response to this strategy of the other agents. If the agent's type is θ and it pretends to be an agent of type $t \in [\underline{\theta}, \overline{\theta}]$, its payoff is

$$\pi_v(1-G(t)) - \theta X(t).$$

Taking the first order condition, we get

$$\pi_v'(1-G(t))(-g(t)) - \theta X'(t) = 0.$$

Now we can plug in $t = \theta$ to get the condition for $X(\theta)$ to be a symmetric Bayes-Nash equilibrium. Doing so, we get

$$\pi_v'(1-G(\theta))(-g(\theta)) - \theta X'(\theta) = 0$$

so that

$$X(\theta) = \int_{\theta}^{\overline{\theta}} \frac{\pi_v'(1 - G(t))g(t)}{t} dt.$$

□

We now state and prove the convergence result.

Theorem 5 (Equilibrium convergence)**.** *Suppose there are $N+1$ agents and fix any contest $v \in \mathcal{V}$. Let $G : [\underline{\theta}, \overline{\theta}] \to [0,1]$ be a differentiable CDF and let $G^1, G^2, \ldots,$ be any sequence of CDF's, each with a finite support, such that for all $\theta \in [\underline{\theta}, \overline{\theta}]$,*

$$\lim_{n\to\infty} G^n(\theta) = G(\theta).$$

Let $F^n : \mathbb{R} \to [0,1]$ denote CDF of the equilibrium effort under G^n, and let $F : \mathbb{R} \to [0,1]$ denote CDF of the equilibrium effort under G. Then, the sequence of CDF's $F^1, F^2, \ldots,$ converges to the CDF F, i.e., for all $x \in \mathbb{R}$,

$$\lim_{n\to\infty} F^n(x) = F(x).$$

Proof. For the finite support CDF G^n, let $\Theta^n = (\theta_1^n, \theta_2^n, \ldots, \theta_{K(n)}^n)$ denote the support and $p^n = (p_1^n, p_2^n, \ldots, p_{K(n)}^n)$ denote the probability mass function. From Theorem 1, let $b^n = (b_1^n, b_2^n, \ldots, b_{K(n)}^n)$ denote the boundary points, $u^n = (u_1^n, u_2^n, \ldots, u_{K(n)}^n)$ denote the equilibrium utilities, and F_k^n denote the equilibrium CDF of agent of type θ_k^n on support $[b_{k-1}^n, b_k^n]$. Then, the CDF of the equilibrium effort, $F^n : \mathbb{R} \to [0,1]$, is such that for any $x \in \mathbb{R}$,

$$F^n(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ P_{k-1}^n + p_k^n F_k^n(x) & \text{if } x \in [b_{k-1}^n, b_k^n] \\ 1 & \text{if } x \geq b_{K(n)}^n \end{cases}. \tag{9}$$

For the continuum CDF $G : [\underline{\theta}, \overline{\theta}] \to [0,1]$, the CDF of the equilibrium effort, $F : \mathbb{R} \to [0,1]$, is such that for any $x \in \mathbb{R}$,

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 - G(\theta(x)) & \text{if } x \in [0, B] \\ 1 & \text{if } x \geq B \end{cases}. \tag{10}$$

where $\theta(x)$ is the inverse of $X(\theta)$ (from Lemma 4) and $B = X(\underline{\theta})$. The following Lemma will be the key to showing that $F^n(x)$ converges to $F(x)$ for all $x \in \mathbb{R}$.

Lemma 5. *Consider any $\theta \in (\underline{\theta}, \overline{\theta})$ and for any $n \in \mathbb{N}$, let $k(n) \in \{0, 1, 2, \dots, K(n)\}$ be such that $\theta^n_{k(n)} > \theta \geq \theta^n_{k(n)+1}$ (where $\theta^n_0 = \infty$ and $\theta^n_{K(n)+1} = 0$). Then,*

$$\lim_{n\to\infty} b^n_{k(n)} = X(\theta) \text{ and } \lim_{n\to\infty} F^n(b^n_{k(n)}) = 1 - G(\theta).$$

Proof. From Lemma 4 and Theorem 1, we have

$$X(\theta) = \int_\theta^{\overline{\theta}} \frac{\pi_v'(1-G(t))g(t)}{t} dt \text{ and } b^n_{k(n)} = \sum_{j=1}^{k(n)} \frac{\pi_v(P^n_j) - \pi_v(P^n_{j-1})}{\theta^n_j}.$$

Observe that

$$\begin{aligned}
b^n_{k(n)} &= \left[\frac{\pi_v(P^n_{k(n)})}{\theta^n_{k(n)}} - \sum_{j=1}^{k(n)-1} \pi_v(P^n_j)\left[\frac{1}{\theta^n_{j+1}} - \frac{1}{\theta^n_j}\right]\right] \\
&= \int_0^{1/\theta^n_{k(n)}} \left[\pi_v(P^n_{k(n)}) - \pi_v(1 - G^n(1/x))\right] dx \\
&\xrightarrow{n\to\infty} \int_0^{\frac{1}{\theta}} \left[\pi_v(1-G(\theta)) - \pi_v(1-G(1/x))\right] dx \qquad \text{(dominated convergence)} \\
&= \underbrace{\left[x(\pi_v(1-G(\theta)) - \pi_v(1-G(1/x)))\right]_0^{\frac{1}{\theta}}}_{\text{this is 0}} + \int_0^{\frac{1}{\theta}} \frac{\pi_v'(1-G(1/x))g(1/x)}{x} dx \\
&= \int_\theta^\infty \frac{\pi_v'(1-G(t))g(t)}{t} \, \mathrm{d}t \qquad \text{(substitute } t = 1/x\text{)} \\
&= X(\theta)
\end{aligned}$$

By definition, we have

$$\begin{aligned}
\lim_{n\to\infty} F^n(b^n_{k(n)}) &= \lim_{n\to\infty} P^n_{k(n)} \\
&= \lim_{n\to\infty} [1 - G^n(\theta)] \\
&= 1 - G(\theta)
\end{aligned}$$

□

Returning to the proof of Theorem 5, fix any $x \in (0, B)$ and let $\theta \in (\underline{\theta}, \overline{\theta})$ be such that $X(\theta) = x$. Then, we know that

$$F(x) = 1 - G(\theta).$$

We want to show that

$$\lim_{n\to\infty} F^n(x) = 1 - G(\theta).$$

Take $\epsilon > 0$. Find $\theta' < \theta$ and $\theta'' > \theta$ such that

$$0 < G(\theta) - G(\theta') = G(\theta'') - G(\theta) < \frac{\epsilon}{4}.$$

Let $x' = X(\theta')$, $x'' = X(\theta'')$, so that $x' > x > x''$. Let $\delta = \min\{x' - x, x - x''\}$. From Lemma 5, let $N(\epsilon)$ be such that for all $n > N(\epsilon)$,

$$\max\{|b^n_{k(n)} - x|, |b^n_{k'(n)} - x'|, |b^n_{k''(n)} - x''|\} < \frac{\delta}{2}$$

and

$$\max\{|F^n(b^n_{k'(n)}) - (1 - G(\theta'))|, |F^n(b^n_{k''(n)}) - (1 - G(\theta''))|\} < \frac{\epsilon}{4},$$

where $k(n), k'(n), k''(n)$ are sequences as defined in Lemma 5 for θ, θ' and θ'' respectively. Then, for all $n > N(\epsilon)$,

$$\begin{aligned} F^n(x) &> F^n(b^n_{k''(n)}) \\ &> 1 - G(\theta'') - \frac{\epsilon}{4} \\ &> 1 - G(\theta) - \frac{\epsilon}{2} \end{aligned}$$

and

$$\begin{aligned} F^n(x) &< F^n(b^n_{k'(n)}) \\ &< 1 - G(\theta') + \frac{\epsilon}{4} \\ &< 1 - G(\theta) + \frac{\epsilon}{2} \end{aligned}$$

so that $|F^n(x) - (1 - G(\theta))| < \epsilon$. Thus, $\lim_{n\to\infty} F^n(x) = 1 - G(\theta) = F(x)$ for all $x \in \mathbb{R}$. □

E Regression tables

Table 3: OLS for Effort Choice as Function of Prizes

	$\theta = 2$	$\theta = 1$	Pooled
First place prize (v_3)	0.401***	0.524***	0.426***
	(0.010)	(0.012)	(0.009)
Second place prize (v_2)	0.321***	0.335***	0.324***
	(0.016)	(0.015)	(0.015)
Third place prize (v_1)	0.314***	0.334***	0.318***
	(0.030)	(0.026)	(0.026)
N	1780	1780	1780
R^2	0.734	0.806	0.786

Standard errors in parentheses clustered at the subject level.

Constant omitted from estimation. $\theta = 2$ is the inefficient type, $\theta = 1$ the efficient type.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 4: OLS for Effort

	Model 1	Model 2
Constant	43.866***	26.087***
	(1.004)	(3.252)
Treatment *Med*	0.472	0.472
	(0.664)	(0.665)
Treatment *High*	2.715***	2.715***
	(0.687)	(0.687)
Treatment *WTA*	7.203***	7.203***
	(0.735)	(0.736)
Inefficient type ($\theta = 2$)	−8.900***	−8.900***
	(0.939)	(0.939)
Male (1 = yes)		−2.285
		(1.542)
Willingness to take risks (0–10)		1.528***
		(0.499)
Willingness to compete (0–10)		1.063**
		(0.503)
N	3560	3560
R^2	0.051	0.106

Standard errors in parentheses clustered at the subject level.

Treatment *Low* is the omitted category. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.